

Dear PLOS ONE Editorial Team,

Please find enclosed our manuscript entitled

"Multi-source domain generalization with few-shot calibration for cross-dataset EEG state classification under proxy labels"

for consideration as a Research Article. This study presents a multi-source domain generalization framework that aligns eight public EEG datasets to a common 14-channel representation and evaluates zero-shot transfer and few-shot calibration for cross-dataset EEG state classification under weak proxy-derived labels; we explicitly refrain from claiming clinical hypnosis-depth classification and instead frame the three-class labels (Awake/Light/Deep) as indicative EEG state categories derived from task conditions, event-phase markers, or arousal self-reports. The work meets PLOS ONE's publication criteria in that the methods are rigorous and reproducible, all quantitative claims are regenerated from a single fixed-seed script with deposited result files, and the conclusions are supported by the data without overstatement (including an explicit caveat that aggregate calibration gain is driven by a single outlying dataset while balanced accuracy remains at chance). We confirm this manuscript is original, not under consideration elsewhere, and approved by all authors; we further declare full compliance with PLOS ONE policies, including an approved IRB exemption (YSUIRB-202607-HR-219-02), a public Data and Code Availability Statement backed by an open GitHub repository, a STROBE/TRIPOD+AI-style reporting checklist provided as Supplementary File S1, a completed PLOS Human Participants Research Checklist provided as Supplementary File S2, and completed Funding, Competing Interests, and Author Contributions (CRediT) statements. We believe this cross-dataset benchmarking study will be of interest to the brain-computer interface, affective computing, and machine-learning reproducibility communities.

Sincerely,

Weng Zexiao

Department of Computer Engineering, Youngsan University

Email: wengzexiao@office.ysu.ac.kr

Jung Minpo (Corresponding Author)

Associate Professor, Department of Computer Engineering, Youngsan University

Email: minpo@ysu.ac.kr